How it breaks

What a KV-value edit actually does when you push it too far

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27 588 generations kept, **10** models (0.5B–8B, six families plus a size ladder inside Qwen2.5), **2** datasets, **1 060** teacher-forced readout points, greedy throughout. Seven bugs in my own instruments, five of which had already produced a finding, and four claims withdrawn — including one headline that reversed twice. Assume there are more.

1. The problem, and the fix that looks right

We change system prompts all the time and nobody deletes the conversation. So transcripts fill with instructions that were right once — *always reply in lowercase* — and then the rule changes and the transcript doesn't. V-Steer (Zeng et al., COLM 2026, arXiv:2607.26228) scales down the cached value vectors of the heads that lean on the stale span, and the old instruction loses its grip.

It works. This note is about what it costs, which meant pushing it until it broke. It did not break the way I expected. It never said *I don't know.*

2. Four answers

> ***Q: What does my account number end in?*** *(gold: 302)*
> *your account number ends in 02.*
>
> ***Q: What is my dog called?*** *(gold: Bagr)*
> *YOUR Dog is called BAGel.*
>
> ***Q: Which city do I live in?*** *(gold: Brno)*
> *you live in a city called Brno is not correct, you said Brno is not mentioned, you said you live in a city that is not mentioned...*
>
> ***Q: What was the error code I got?*** *(gold: E-88)*
> *You mentioned earlier that the error code was E-88, but I'm not sure if that's correct, you actually said it was E-88 but then mentioned it was E-88 again and then said E-88 was actually E-88 but then said...*

Fluent, correctly cased, obedient to the current system prompt, wrong. The third one is the one to worry about: the answer contains the right city and refuses to admit anyone said it. Count correct strings — success. Count rule violations — also success. Two standard metrics, both green.

3. One mechanism

Teacher-force both conditions to the exact position where the value is emitted, so the same question is asked in the same place, and compare the next-token distributions. On all ten models: **the correct token is not overwritten, it is attenuated.** Nothing promotes a wrong token; the right one is pushed down until whatever stood behind it walks forward.

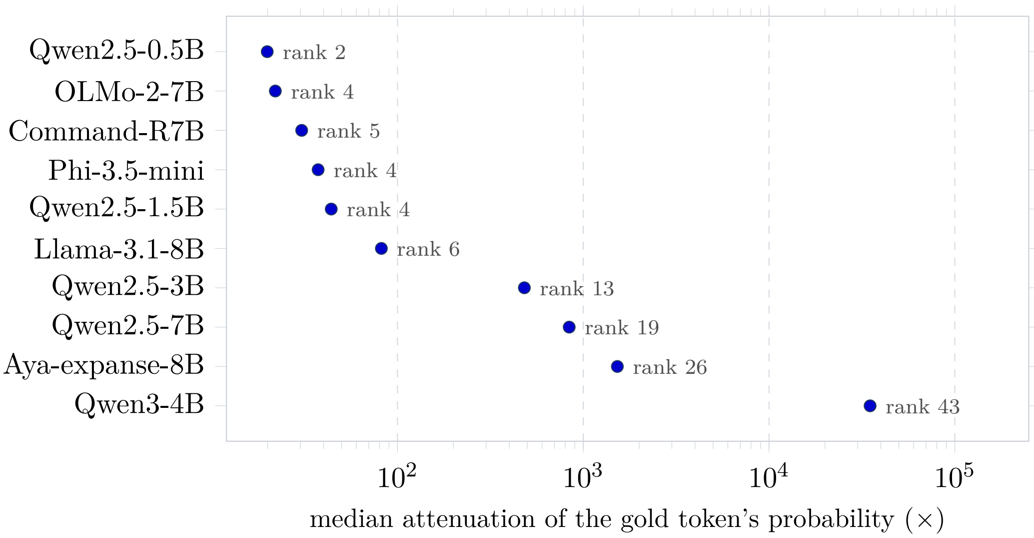

Figure 1: Every model attenuates; the magnitude spans more than three orders of magnitude (1 760× between the mildest model and the harshest). The annotation is the rank the substituted token held in the *unsteered* distribution — on the small models the edit hands the slot to the runner-up, on Qwen3-4B it reaches down to rank 43.

The direction is universal and the size is not. The gold token's own rank stays in single digits even as its probability falls by four orders — still near the top, just no longer on it. That is what scaling V while leaving attention alone predicts: the model looks straight at the fact and receives a whisper.

4. The modes, and how each is measured

Near neighbour. `302`→`02` (32×), `bagr`→`Bagel` (17), `4417-B`→`4411` (15) on Llama. Scoring each wrong answer against a gold value from a *different* question gives 0–2 % against 68 % for its own, so the replacement is specifically related to the target. But the *rate* is mostly the definition: 9 % under a shape-only rule, 68 % once truncation counts. "Near misses dominate" is not a claim this data supports.

States the fact, denies being told it. By rule — the string is present *and* a denial pattern fires — Llama 16/756, three models 1/756, six models zero. Dose-dependent on Llama (0, 2, 1, 6, 7 as γ_- rises).

Non-terminating recall. The *correct* value three to six times, never settling. Llama 15/756, Qwen3-4B 2/756, zero on the other eight.

Format kept, confabulation poured in. Where neither rule wins on Qwen3-4B, 64 of 119 cases recalled the fact correctly and broke the format instead — median 38 words against 17 — filled with material nobody supplied (*"Bagr… perhaps inspired by the ancient Persian word for 'to be strong'"*).

Both odd modes are deterministic rules, not judge labels. The judge reproduces at 94–100 % on the easy categories and 58 % on the newest one, so it does not get to decide the interesting ones — its 26 unsourced labels on Aya are false positives, and its 217 unresolved labels on Phi are that model's ordinary repetition, which is *highest with no steering at all.*

5. What is universal, and what to actually ship

Universal: attenuation. Not universal, and not a size effect: which mode you get. The two odd modes are Llama's almost exclusively, while OLMo, Aya and Command-R are the same size and never do it, and the whole Qwen2.5 ladder produces no loops at all.

The practical consequence is the reason any of this matters. **Every mode here is invisible to the metric you would normally use.** A violation counter scores the denial as a miss and the confabulation as a success; a needle-presence check scores the denial and the loop as successes.

And the least glamorous result, which is the one I would ship: where the stale instruction sits in its own message, **deleting that message beats the method on 7 of 10 models**. The edit earns its place only when the instruction is entangled with something you still need — and even there, rewriting the message beats it on the strong models. It wins outright on a model too weak to follow the current rule unaided, which is a narrower claim than the one I went looking for.

Constructed cases, six facts, seven constraint families, English, greedy, one reimplementation of one method. Reproduce the mode table with `python examples/mode_rates.py results/atlas_*.rescored.json` — no GPU, no API. Raw generations for all of it are in `results/`.